Holomorphic Functions

1 Definition

A function $f : U \subset \mathbb{C} \to \mathbb{C}$ is called **holomorphic** on an open set U if it is complex differentiable at every point in U. More formally, f is holomorphic if for every point $z_0 \in U$, the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \tag{1}$$

exists. This limit defines the derivative $f'(z_0)$. A function is holomorphic in a neighborhood of every point in its domain, meaning it is complex differentiable in some open disk around each point.

Holomorphic functions satisfy the Cauchy-Riemann equations, which are necessary and sufficient conditions for differentiability in the complex sense. If $f(z) = u(x, y) + iv(x, y)$ where $z = x + iy$, then the Cauchy-Riemann equations state:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{2}$$

2 Examples

1. **Polynomial Functions**: Any polynomial $f(z) = a_n z^n + a_{n-1} z^{n-1} + \ldots + a_1 z + a_0$ where $a_i \in \mathbb{C}$ is holomorphic everywhere in $\mathbb{C}$.

2. **Exponential Function**: The function $f(z) = e^z$ is holomorphic everywhere in $\mathbb{C}$. Its derivative is $f'(z) = e^z$, which is also holomorphic.

3. **Trigonometric Functions**: The sine and cosine functions, defined as $f(z) = \sin(z)$ and $g(z) = \cos(z)$, are holomorphic everywhere in $\mathbb{C}$.

4. **Rational Functions**: The function $f(z) = \frac{1}{z}$ is holomorphic on $\mathbb{C} \setminus \{0\}$ (it is not defined at $z = 0$).

5. **Complex Logarithm**: The function $f(z) = \log(z)$ is holomorphic on $\mathbb{C} \setminus \{0\}$ but has a branch cut, typically along the negative real axis.

3 Key Results

- **Cauchy's Integral Theorem**: If f is holomorphic on a simply connected domain D, then for any closed contour C in D:

$$\oint_C f(z)\, dz = 0. \tag{3}$$

- **Cauchy's Integral Formula**: If f is holomorphic inside and on some simple closed contour C, then for any point a inside C:

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a}\, dz. \tag{4}$$

- **Liouville's Theorem**: A bounded entire function (holomorphic on all of $\mathbb{C}$) must be constant.

- **Maximum Modulus Principle**: If f is holomorphic in a domain D, then the maximum of $|f(z)|$ occurs on the boundary of D.

4 Diagrams / Visual Description

Holomorphic functions can be visualized in the complex plane. The mapping of a holomorphic function can be represented by arrows indicating the transformation of points in the domain to points in the codomain. For example, the function $f(z) = z^2$ maps circles centered at the origin to circles in the w-plane, but the radius of these circles changes, demonstrating the local behavior of holomorphic functions.

5 Connections

Holomorphic functions are closely related to several important concepts in complex analysis:

- **Cauchy-Riemann Equations**: These equations provide the necessary conditions for a function to be holomorphic, linking real and imaginary parts of complex functions.

- **Burgers Transform**: This is a transformation that relates holomorphic functions to solutions of certain partial differential equations, emphasizing the interplay between complex analysis and fluid dynamics.

- **C1 Functions**: Holomorphic functions are infinitely differentiable (smooth), which implies they are C^1 functions.

- **Obstruction Formula**: The study of holomorphic functions often involves understanding obstructions to extending holomorphic structures, particularly in complex geometry.

6 References

- Alayón-Solarz, D. (2023). "Variable Elliptic Structures on the Plane: Definitions." arXiv:2601.19274v4.
- Alayón-Solarz, D. (2023). "Covariantly Holomorphic Functions and the Cauchy–Pompeiu Formula." arXiv:2602.19251v1.